

Chapter 5: Excitations

14.0 Release

Fluid Dynamics

Structural Mechanics

Electromagnetics

Systems and Multiphysics

Introduction to ANSYS HFSS

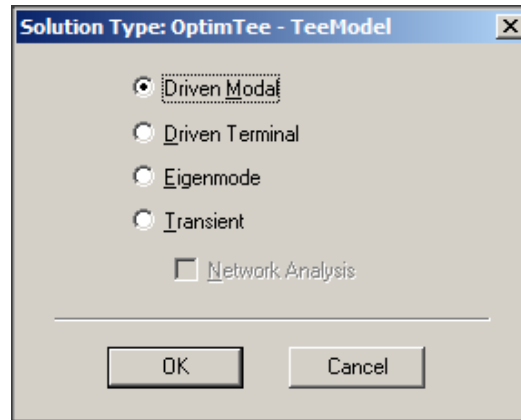
- **Ports**

- Ports are a unique type of boundary condition that allows energy to flow into and out of a structure. You can assign a port to any 2D object or 3D object face. Before the full three-dimensional electromagnetic field inside a structure can be calculated, it is necessary to determine the excitation field pattern at each port. HFSS uses an arbitrary port solver to calculate the natural field patterns or modes that can exist inside a transmission structure with the same cross section as the port. The resulting 2D field patterns serve as boundary conditions for the full three-dimensional problem.
- By default HFSS assumes that all structures are completely encased in a conductive shield with no energy propagating through it. You apply **Wave Ports** to the structure to indicate the area where the energy enters and exits the conductive shield.
- As an alternative to using Wave Ports, you can apply **Lumped Ports** to a structure instead. Lumped Ports are useful for modeling internal ports within a structure.
- **Network Parameters**
 - HFSS uses the incident and reflected energy at the port to compute Generalized S-Parameters
 - Generalized S-parameters normalize the S-parameters by the port's (transmission line's) characteristic impedance
 - Y and Z parameters are then derived from these generalized S-parameters
 - Ports are the only excitations that can be used to compute Network Parameters (S, Y and Z Parameters)

Setting Solution Type

- **Solution type selection**

- As a general rule, one could choose the solution type based on the type of transmission line that is being analyzed
- **Driven Modal**
 - Hollow waveguides (metallic rectangular, circular...etc)
 - Any problem where a symmetry boundary condition is applied
- **Driven Terminal**
 - Microstrip, stripline, coax, coplanar waveguide



HFSS Excitation Methods

Driven Modal

- Fields based transmission line interpretation
- Port's signal decomposed into incident and reflected waves
- Excitation's magnitude described as an incident power

Modal Propagation

- Energy propagates in a set of orthogonal modes
- Modes can be TE, TM and TEM w.r.t. the port's normal
- Mode's field pattern determined from entire port geometry
- Each Mode has its own column and row in the S, Y, and Z parameters



Driven Terminal

- Circuit Based transmission line interpretation
- Port's signal interpreted as a total voltage ($V_{total} = V_{inc} + V_{ref}$)
- Excitation's magnitude described as either a total voltage or an incident voltage
- Supports Differential S-Parameters

Terminal Propagation

- Each conductor touching the port is considered a terminal or a ground
- Energy propagates along each terminal in a single TEM mode
- Each Terminal has its own column and row in the S, Y and Z parameters
- Does not support symmetry boundaries or Floquet Ports



- **Wave Port**

- The port solver assumes that the Wave Port you define is connected to a semi-infinitely long waveguide that has the same cross-section and material properties as the port. Each Wave Port is excited individually and each mode incident on a port contains one watt of time-averaged power. Wave Ports calculate characteristic impedance, complex propagation constant, and generalized S-Parameters.

- **Wave Equation**

- The field pattern of a traveling wave inside a waveguide can be determined by solving Maxwell's equations. The following equation that is solved by the 2D solver is derived directly from Maxwell's equation.

$$\nabla \times \left(\frac{1}{\mu_r} \nabla \times E(x, y) \right) - k_0^2 \epsilon_r E(x, y) = 0$$

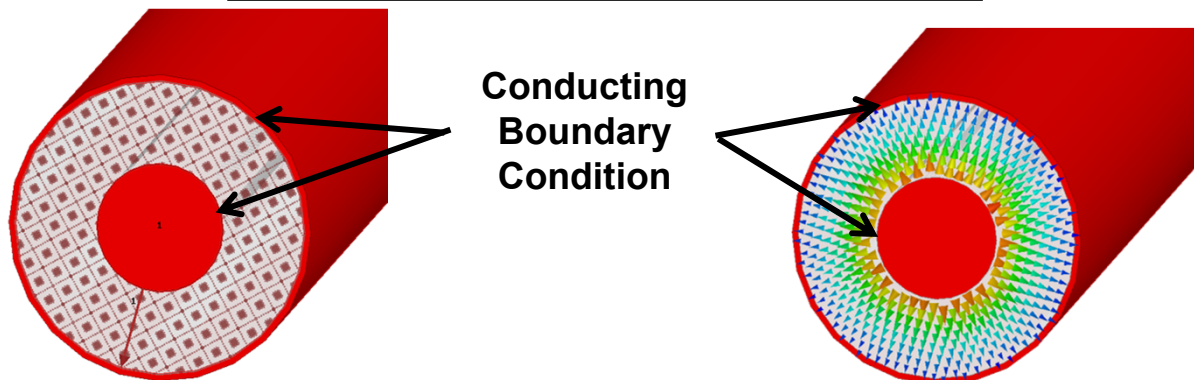
- where:
 - $E(x, y)$ is a phasor representing an oscillating electric field.
 - k_0 is the free space wave number,
 - μ_r is the complex relative permeability.
 - ϵ_r is the complex relative permittivity.
- To solve this equation, the 2D solver obtains an excitation field pattern in the form of a phasor solution, $E(x, y)$. These phasor solutions are independent of z and t ; only after being multiplied by $e^{-\gamma z}$ do they become traveling waves.
- Also note that the excitation field pattern computed is valid only at a single frequency. A different excitation field pattern is computed for each frequency point of interest.

Wave Ports

- **External port type**

- Arbitrary port solver calculates natural waveguide field patterns (modes)
 - Assumes semi-infinitely long waveguide with same cross-section and material properties as port surface
- Recommended only for surfaces exposed to background object
- Supports multiple modes, de-embedding, and re-normalization
- Computes generalized S-parameters
 - Frequency-dependent characteristic impedance
 - Perfectly matched at every frequency
- HFSS places a boundary condition on the Wave Port's edges to make sure the 2D eigenmode solution is finite sized
 - Typically the edge boundary takes the boundary condition defined in the 3D geometry.
 - EXCEPTION: When a Wave Port's edge touches a radiation boundary the edge's boundary is defined as PEC.
 - The Wave Port must encompass all relevant E-fields needed to describe the transmission line's behavior.

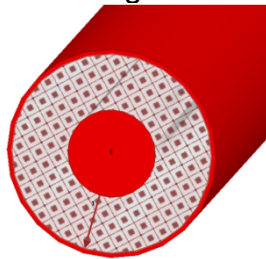
Freq	Port Zo	Gamma (Np/meter, rad/meter)	Lambda (meters)	Epsilon
10 (GHz)	1:1 (50.218, 0)	(0, 209.58)	0.029979	1
	2:1 (50.553, 0)	(0, 209.58)	0.029979	1



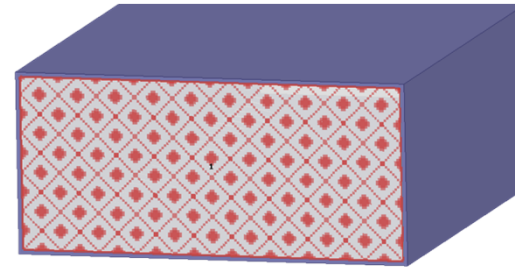
Wave Port Sizing

- **Closed Transmission Line Structures**

- The boundary enforced on the port's edge implies the transmission line modeled by the Wave Port always sits inside a waveguide structure
- The enclosing material forms the port's edge boundary.



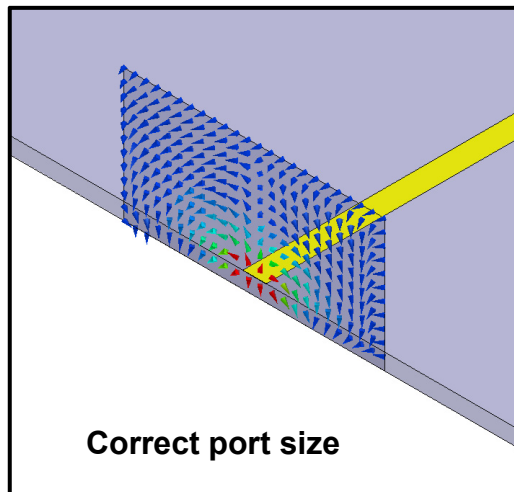
Coax



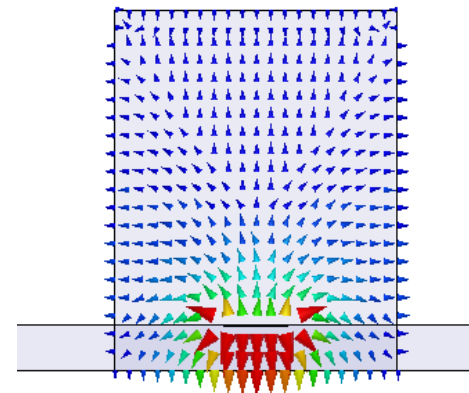
Waveguide

- **Open transmission line structures require additional consideration**

- Microstrip, Co-Planar Waveguide, Slotline (See Appendix for sizing recommendations)
- Make sure the transmission line fields are not interacting with the port's boundary condition.
 - Can lead to incorrect characteristic impedances, will add additional reflection based solely on the port



Correct port size



Port too narrow
(fields coupled to sidewalls)

- **Modes**

- For a waveguide or transmission line with a given cross section, there is a series of basic field patterns (modes) that satisfy Maxwell's Equations at a specific frequency. Any linear combination of these modes can exist in the waveguide.
- **Mode Conversion**
 - In some cases it is necessary to include the effects of higher-order modes because the structure acts as a mode converter. For example, if the mode 1 (dominant) field at one port is converted (as it passes through a structure) to a mode 2 field pattern at another, then it is necessary to obtain the S-parameters for the mode 2 field.
- **Modes, Reflections, and Propagation**
 - It is also possible for a 3D field solution generated by an excitation signal of one specific mode to contain reflections of higher-order modes which arise due to discontinuities in a high frequency structure. If these higher-order modes are reflected back to the excitation port or transmitted onto another port, the S-parameters associated with these modes should be calculated. If the higher-order mode decays before reaching any port—either because of attenuation due to losses or because it is a non-propagating evanescent mode—there is no need to obtain the S-parameters for that mode.
- **Modes and Frequency**
 - The field patterns associated with each mode generally vary with frequency. However, the propagation constants and impedances always vary with frequency. Therefore, when a frequency sweep has been requested, a solution is calculated for each frequency point of interest. When performing frequency sweeps, be aware that as the frequency increases, the likelihood of higher-order modes propagating also increases.

Wave Port Boundary Conditions

- **Wave Port Boundary Condition**

- The edge of a Wave Port can have the following boundary conditions:
 - **Perfect E or Finite Conductivity** – by default the outer edge of a Wave Port is defined to have a Perfect E boundary. With this assumption, the port is defined within a waveguide. For transmission line structures that are enclosed by metal, this is not a problem. For unbalanced or non-enclosed lines, the fields in the surrounding dielectric must be included. Improper sizing of the port definition will result in erroneous results.
 - **Symmetry** – the port solver understands Perfect E and Perfect H symmetry planes. The proper Wave Port impedance multiplier needs to be applied when using symmetry planes.
 - **Impedance** – the port solver will recognize an impedance boundary at the edges of the ports.
 - **Radiation** – the default setting for the interface between a Wave Port and a Radiation boundary is to apply a Perfect E boundary to the edge of the ports.

Considerations for Defining Wave Ports

- **Considerations for Defining Wave Ports**

- **Wave Port Locations**

- It is recommended that only surfaces that are exposed to the background be defined as Wave Ports. The background is given the boundary name outer. Therefore a surface is exposed to the background if it touches the boundary outer. You can locate all regions of outer by selecting the menu item HFSS, Boundary Display (Solver View). From the Solver View of Boundaries, check the Visibility for outer.

- **Interior Wave Ports**

- If you want to apply Wave Ports to the interior of a structure, you must create an inner void or select the surface of an interior object that is assign a perfect conductor material property. Inner voids are automatically assigned the boundary outer. You can create an inner void by surrounding one object entirely with another object, then subtracting the interior object.

- **Ports are Planar**

- A port must lie in a single plane. Ports that bend are not allowed. For example, if a geometric model has a curved surface exposed to the background, that curved surface cannot be defined as a port.

- **Wave Ports Require a Length of Uniform Cross Section**

- HFSS assumes that each port you define is connected to a semi-infinitely long waveguide that has the same cross section as the Wave Port. When solving for S-parameters, the simulator assumes that the structure is excited by the natural field patterns (modes) associated with these cross sections. The following figures illustrate cross sections. The first figure shows regions that have been defined as Wave Ports on the outer conductive surface of a structure.

- **Wave Ports and Multiple Propagating Modes**

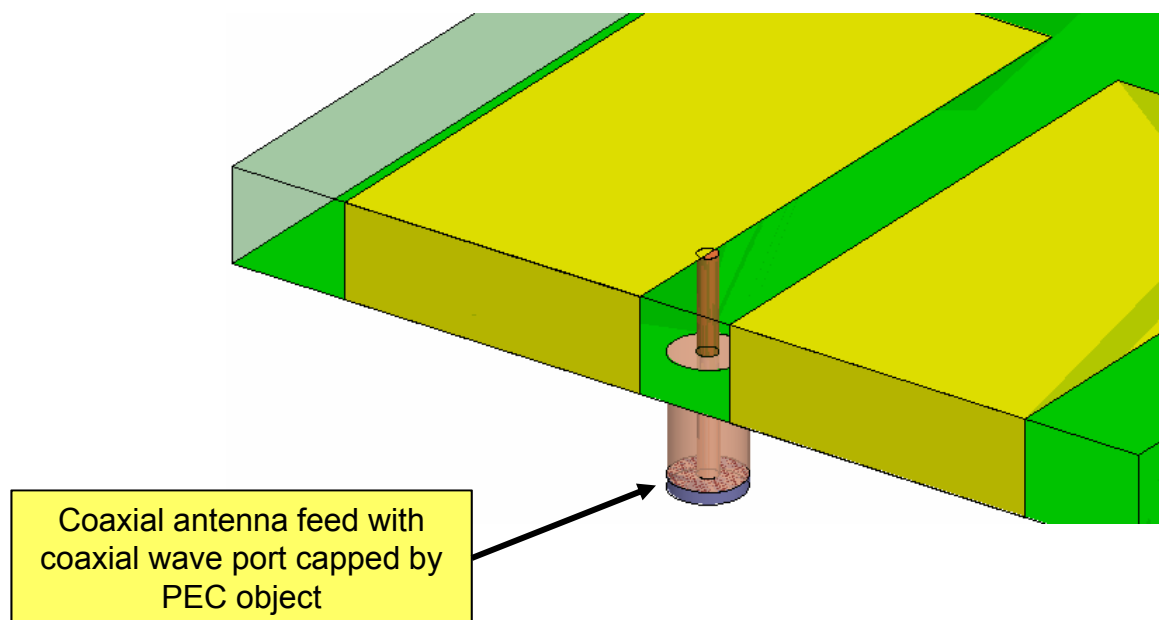
- Each higher-order mode represents a different field pattern that can propagate down a waveguide. In general, all propagating modes should be included in a simulation. In most cases, you can accept the default of 1 mode, but where propagating higher-order modes are present you need to change this to include higher-order modes. If there are more propagating modes than the number specified, erroneous results will be generated. The number of modes can vary among ports.

Internal Wave Ports

- **Internal Wave Ports**

- Wave ports can be placed internal to model by providing boundary condition normally seen by external wave port
 - Create PEC “cap” to back the wave port and enable excitation in proper direction

Example coax feed within solution volume



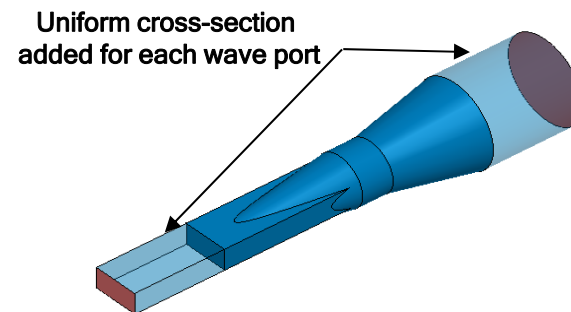
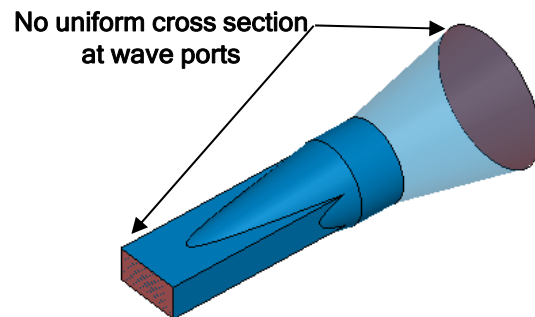
Wave Port Implications

- **Modes, reflections, and propagation**

- It is possible for 3D field solution generated by excitation signal of one specific mode to contain reflections of higher-order modes which arise due to discontinuities
- If higher-order mode is reflected back to excitation port or transmitted onto another port, its S-parameters should be calculated
- If higher-order mode decays before reaching any port (because of attenuation or because it is a non-propagating evanescent mode), there is no need to obtain its S-parameters

- **Wave ports require a length of uniform cross-section**

- HFSS assumes that each port is connected to semi-infinitely long waveguide with same cross-section as wave port



Generalized S-Parameters

- **Modes and S-Parameters**

- When the Wave Ports are defined correctly, for the modes that are included in the simulation, there is a perfect matched condition at the Wave Port. Because of this, the S-Parameters for each mode and Wave Port are normalized to a frequency dependent impedance. This type of S-Parameter is referred to as Generalized S-Parameter.
- Laboratory measurements, such as those from a vector network analyzer, or circuit simulators use a constant reference impedance (i.e. the ports are not perfectly matched at every frequency).
 - To obtain results consistent with measurements or for use with circuit simulators, the generalized s-parameters calculated by HFSS must be renormalized to a constant characteristic impedance. See the section on Calibrating Wave Ports for details on how to perform the renormalization.
 - Note: Failure to renormalize the generalized S-Parameters may result in inconsistent results. For example, since the Wave Ports are perfectly matched at every frequency, the S-Parameters do not exhibit the interaction that actually exists between ports with a constant characteristic impedance.

Generalized S-Parameters

- HFSS normalizes the S-Parameters by the mode's characteristic impedance

$$S_{1,1} = \frac{V_1^-}{V_1^+} \frac{\sqrt{Z_{0,1}}}{\sqrt{Z_{0,1}}} \bigg|_{V_k^+ = 0 \text{ for } k \neq 1} \quad S_{2,1} = \frac{V_2^-}{V_1^+} \frac{\sqrt{Z_{0,1}}}{\sqrt{Z_{0,2}}} \bigg|_{V_k^+ = 0 \text{ for } k \neq 1} \quad \Rightarrow \quad S_{m,n} = \frac{V_m^-}{V_n^+} \frac{\sqrt{Z_{0,n}}}{\sqrt{Z_{0,m}}} \bigg|_{V_k^+ = 0 \text{ for } k \neq n}$$



- **Impact:**

1. Makes every port appear as an infinitely long transmission line over all frequencies
 - Terminates ports in matching transmission line eliminating additional reflections
2. Maintains S-parameter interpretation when different modes have different characteristic impedances
 - Preserves conservation of energy in the S-parameters when mode's have different characteristic impedances.

Calibrating Wave Ports

- **Calibrating Wave Ports**

- Wave Ports that are added to a structure must be calibrated to ensure consistent results. This calibration is required in order to determine direction and polarity of fields and to make voltage calculations.
- **Solution Type: Driven Modal**
 - For Driven Modal simulations, the Wave Ports are calibrated using Integration Lines. Each Integration Line is used to calculate the following characteristics:
 - **Impedance** – As an impedance line, the line serves as the path over which HFSS integrates the E-field to obtain the voltage at a Wave Port. HFSS uses the voltage to compute the characteristic impedance of the Wave Ports, which is needed to renormalize generalized S-matrices to specific impedances such as 50 ohms.
 - **Note:** If you want to be able to renormalize S-parameters or view the values of Zpv or Zvi, you must apply Integration Lines to the Wave Ports of a structure.
 - **Calibration** – As a calibration line, the line explicitly defines the up or positive direction at each Wave Port. At any Wave Port, the direction of the field at $\omega t = 0$ can be in at least one of two directions. At some ports, such as circular ports, there can be more than two possible directions, and you will want to use Polarize E-Field. If you do not define an Integration Line, the resulting S-parameters can be out of phase with what you expect.
 - **Tip** You may need to run a ports-only solution first to help determine how the Integration Lines need to be applied to a Wave Port and their direction.

- **Driven Modal: Calibration**

- **Impedance Lines**

- The S-matrices initially calculated by HFSS are generalized S-matrices normalized to the impedances of each port. However, it is often desirable to compute S-matrices that are normalized to specific impedances such as 50 ohms. To convert a generalized S-matrix to a renormalized S-matrix, HFSS first computes the characteristic impedance at each port. There are several ways to compute the characteristic impedance (Z_{pi} , Z_{pv} , Z_{vi}).
 - HFSS will always calculate Z_{pi} . The impedance calculation using power and current is well defined for a Wave Port. The other two methods – Z_{pv} and Z_{vi} – require a line of integration to determine voltage. By defining an Integration Line for each mode, the voltage can be computed.
 - In general, the impedance line should be defined between two points at which the voltage differential is expected to be a maximum. If you are analyzing multiple modes, define a separate Integration Lines for each mode since the orientation of the electric field will vary.

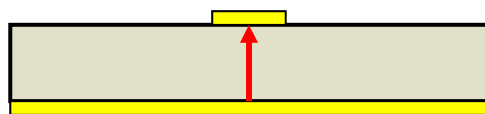
- **Aligning Modes**

- When the excitation field pattern at a Wave Port is computed, the direction of the field at $\omega t=0$ is arbitrary and can point in one of at least two ways. The Integration Lines calibrate the port by defining the preferred direction or the reference orientation. Be sure to define Integration Lines for each Wave Port so that the preferred direction is the same relative to other ports having identical or similar cross-sections. In this way, the results of laboratory measurements (in which the setup is calibrated by removing the structure and connecting two ports together) can be duplicated.
 - Because the lines only determine the phase of the excitation signal and the traveling wave, the system ignores them during the Ports-Only solution

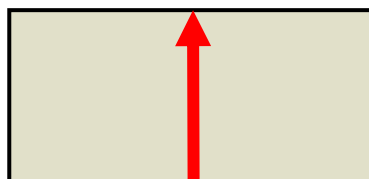
Integration Lines

• Integration Lines

- Applicable to driven modal solution types
- Port vector which can serve several purposes
 - Calibration line which specifies direction of excitation electric field pattern at port
 - Define separate integration line for each mode on multi-mode ports
- Impedance line along which to compute Z_{pv} or Z_{vi} port impedance
 - Select two points with maximum voltage differential



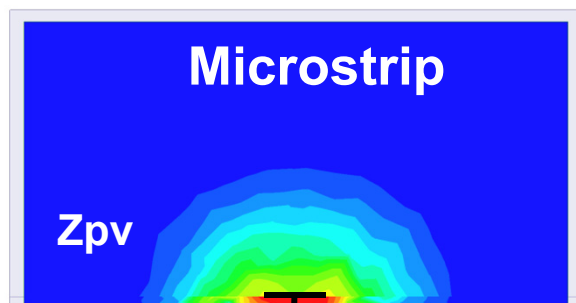
Microstrip line



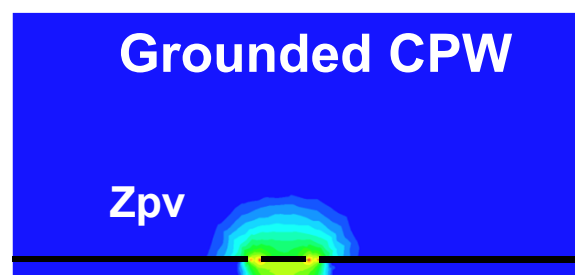
Waveguide



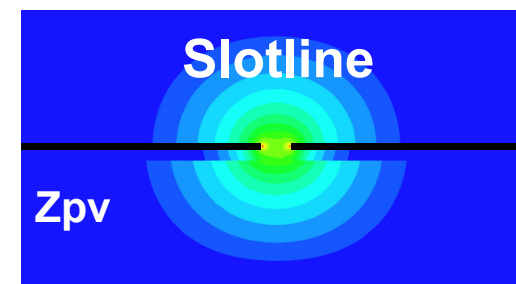
Slotline



Z_{pv} with Integration Line between trace and ground



Z_{pv} with Integration Line between trace and ground



Z_{pv} with Integration Line between ground planes

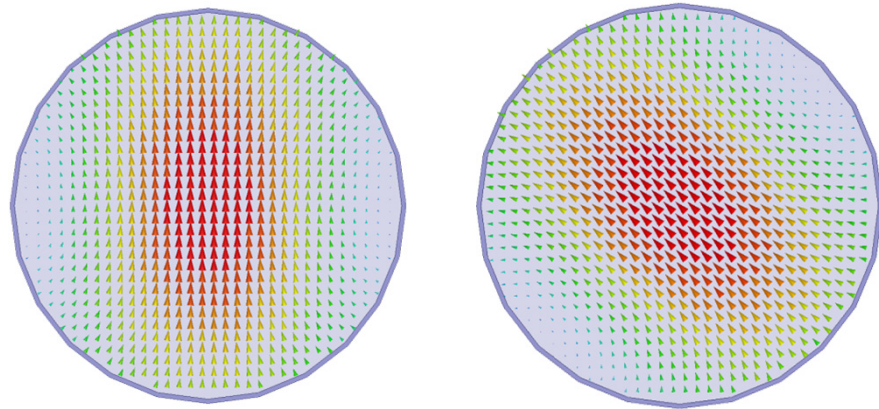
Mode Alignment

- **Mode Alignment**

- Controls orientation of field vectors
- Differentiates unique solutions for degenerate modes

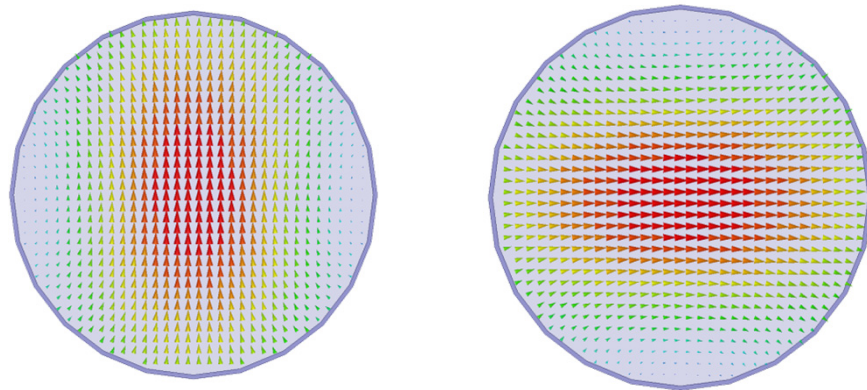
Rotational symmetry produce ambiguity in mode alignment

What direction should the mode point?



Which is mode 1?

Degenerate mode produces ambiguity in mode order

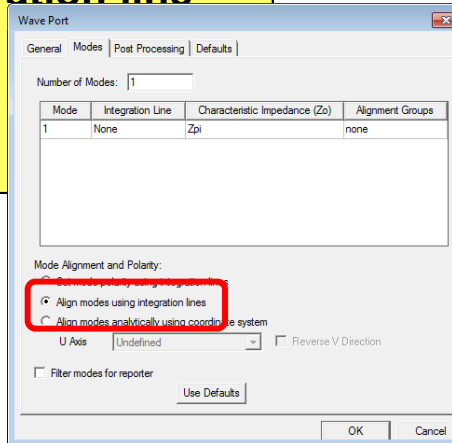


Aligning the Modes

- 2 methods for aligning modes

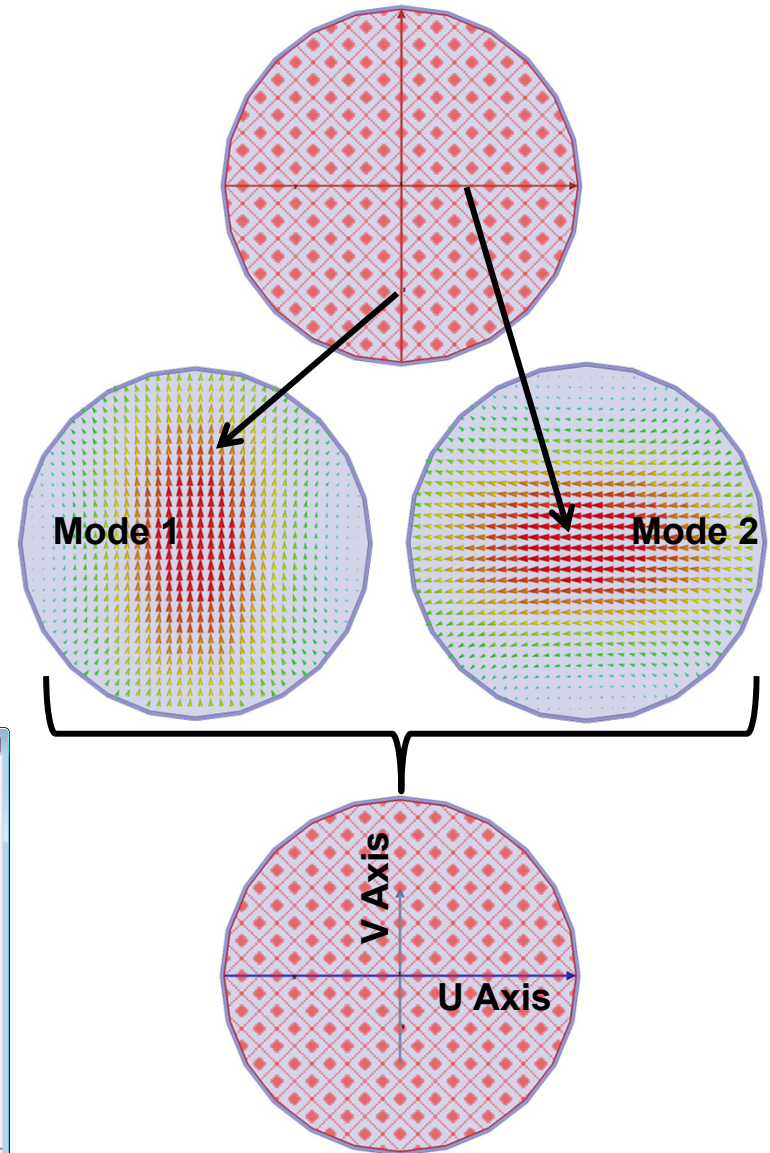
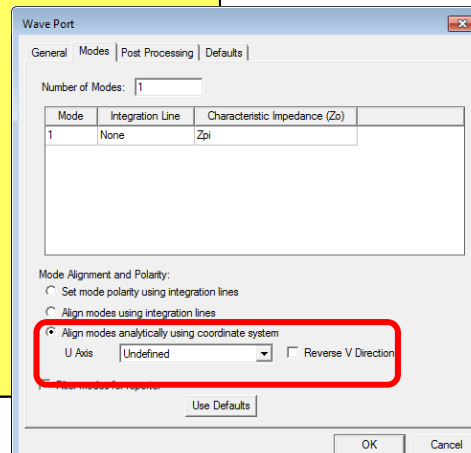
Align modes using integration line

- Uses the integration line to
- align the modes
- Alignment groups help sort out degenerate modes



Align modes analytically using coordinate system

- Uses a locally defined UV coordinate system to align modes
- Only applies to:
 - Coaxial Transmission Line
 - Circular Waveguide
 - Rectangular Waveguide
- U-Axis defined as vector
 - Specify vector tale
 - Specify vector tip



Driven Terminal

- **Calibrating Wave Ports**

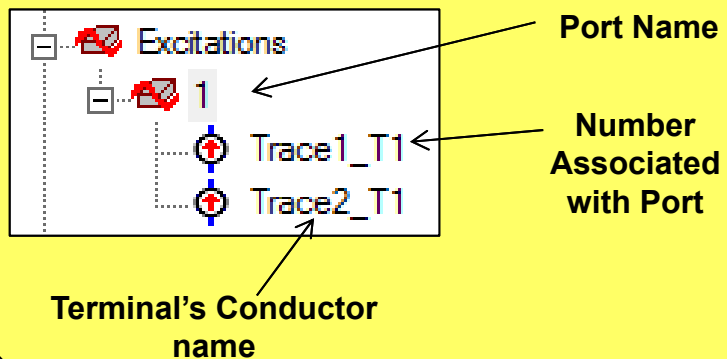
- **Solution Type: Driven Terminal**

- The Modal S-matrix solution computed by HFSS is expressed in terms of the incident and reflected powers of the waveguide modes. This description does not lend itself to problems where several different quasi-transverse electromagnetic (TEM) modes can propagate simultaneously. For structures like coupled transmission lines or connectors, which support multiple, quasi-TEM modes of propagation, it is often desirable to have HFSS compute the Terminal S-Parameters.

Terminal Naming

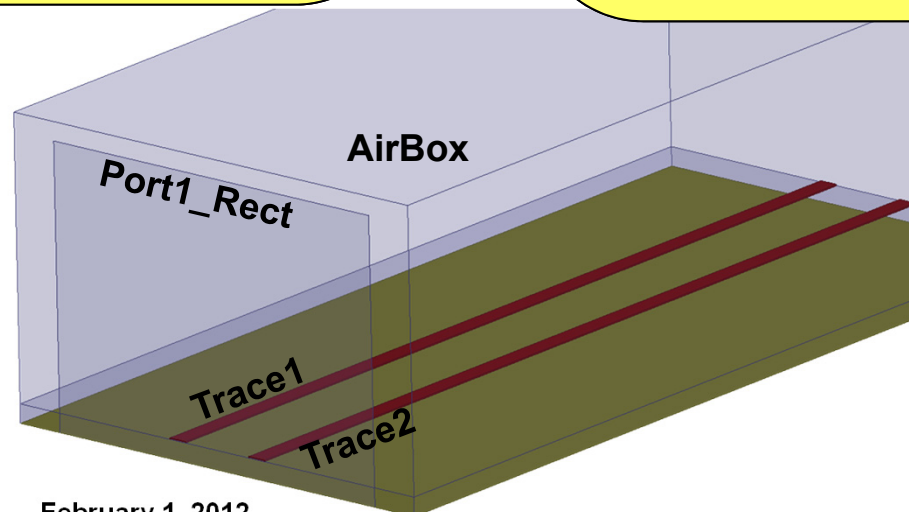
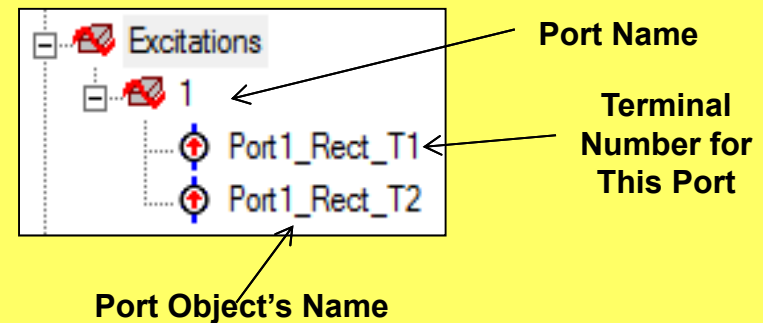
Conductor Name

- Names terminal based on conductor it is assigned



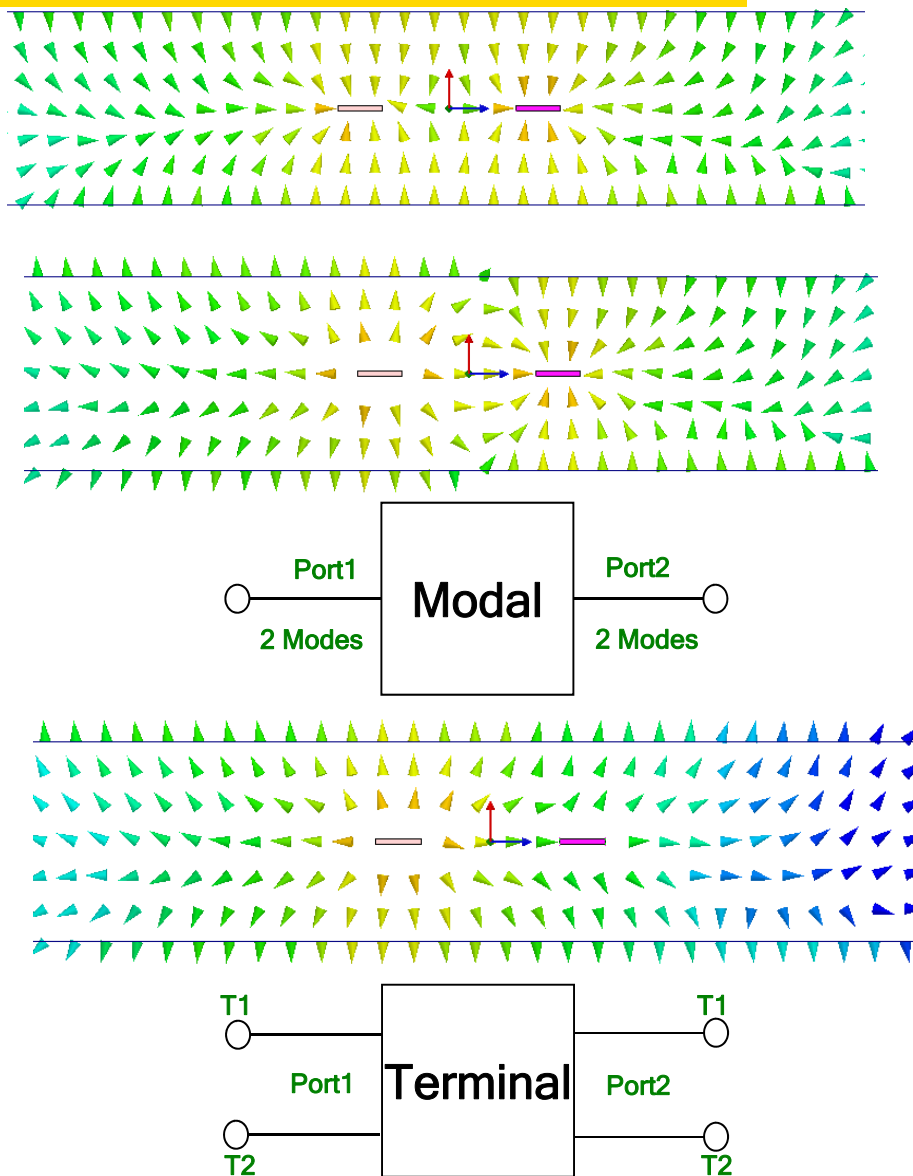
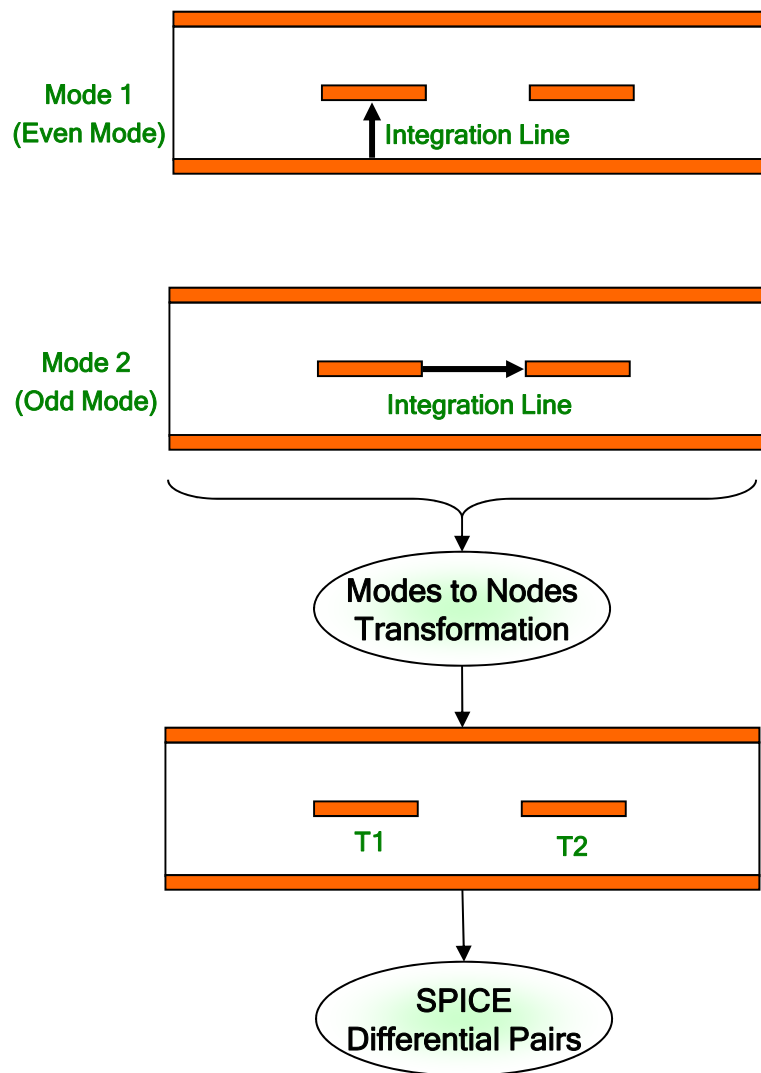
Port Object Name

- Names terminal based on port object's name



Excitations: Modal vs. Terminal

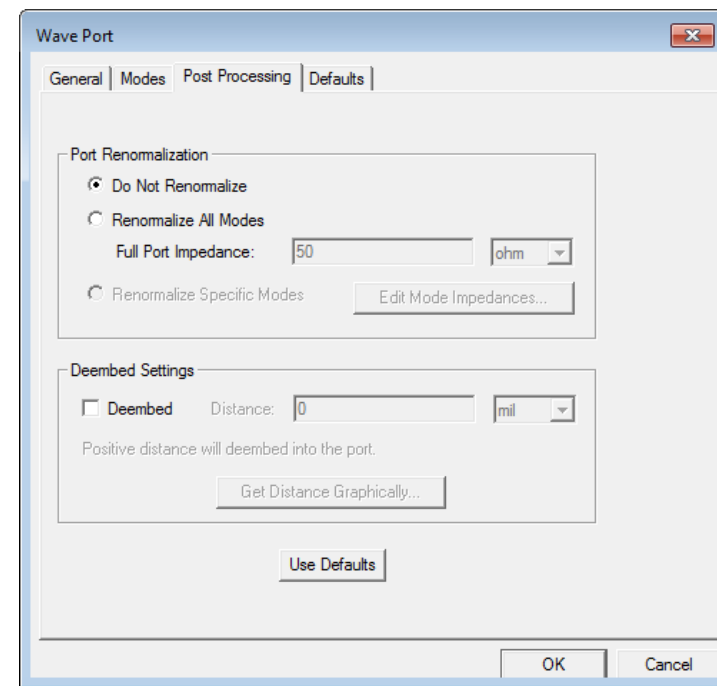
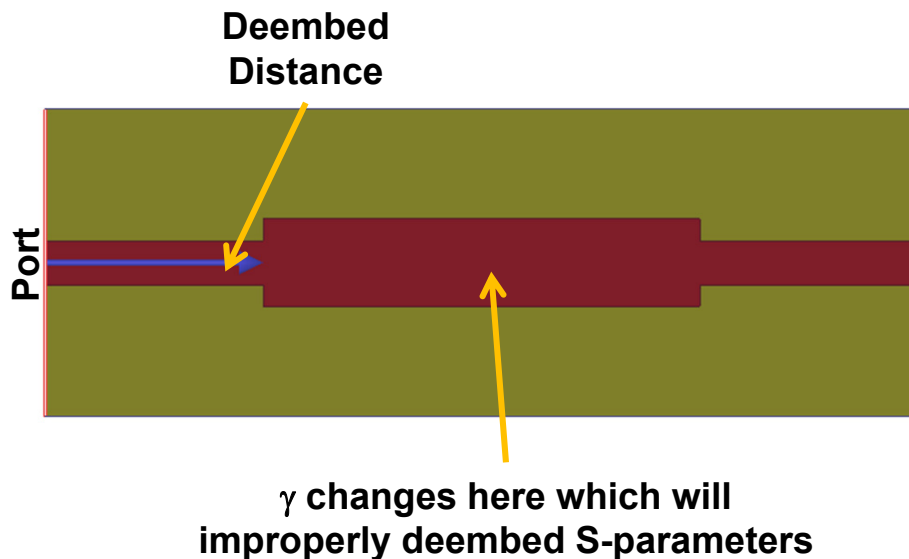
- Example Solution Types:



Wave Port: Post Processing

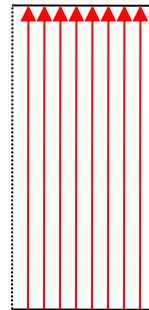
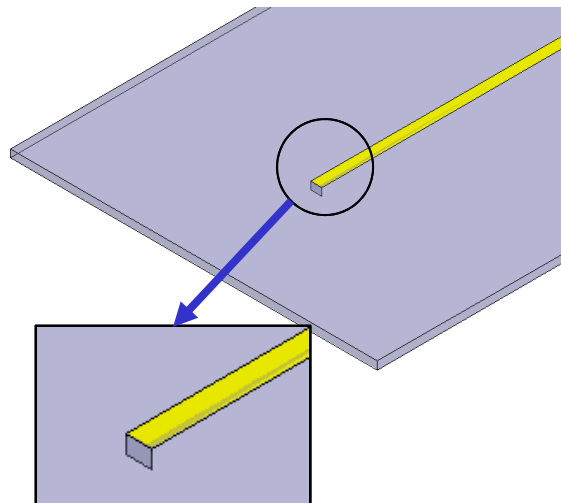
- **Allows re-normalization of the S-parameters to specific characteristic impedances**
 - Required when comparing HFSS results with measured data.
- **Deembeds a uniform length of transmission line**
 - Adjusts S-parameters magnitude and phase based on complex propagation constant calculated at port
 - DO NOT deembed beyond a discontinuity

$$[S^n] = \begin{bmatrix} e^{\gamma_1 \ell_1} & 0 & 0 \\ 0 & e^{\gamma_2 \ell_2} & 0 \\ 0 & 0 & e^{\gamma_n \ell_n} \end{bmatrix} [S] \begin{bmatrix} e^{\gamma_1 \ell_1} & 0 & 0 \\ 0 & e^{\gamma_2 \ell_2} & 0 \\ 0 & 0 & e^{\gamma_n \ell_n} \end{bmatrix}$$

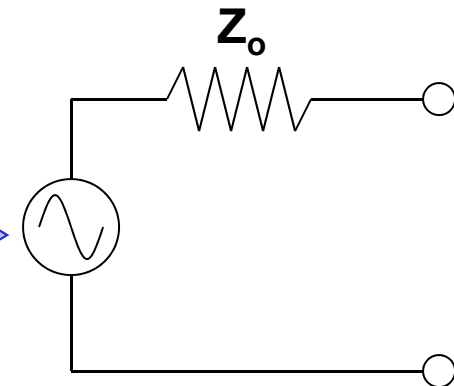


Lumped Ports

- **Recommended only for surfaces internal to model**
 - Single TEM mode with no de-embedding
 - Uniform electric field on port surface
 - Normalized to constant user-defined Z_0
- **Lumped port boundary conditions**
 - Perfect E or finite conductivity boundary for port edges which interface with conductor or another port edge
 - Perfect H for all remaining port edges



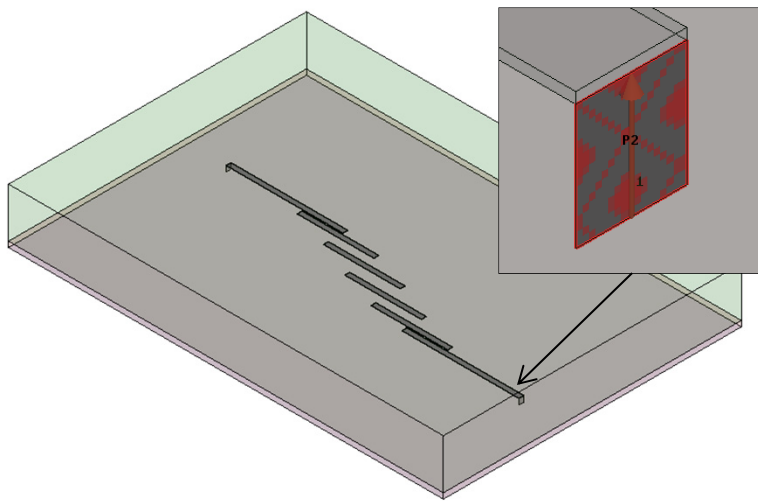
Uniform electric field
User-defined Z_0



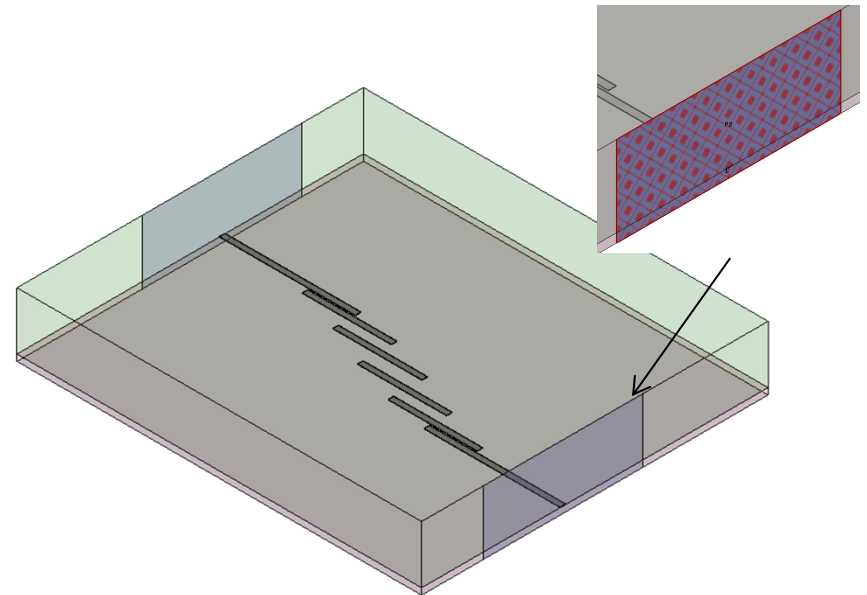
Dipole element
with lumped port

Lumped vs Wave Ports for Planar Filters

- Lumped ports can be used to feed printed transmission lines
 - S-parameters normalized to user-specified characteristic impedance
 - Single mode propagation
 - No de-embedding operations available
 - Must be located inside model

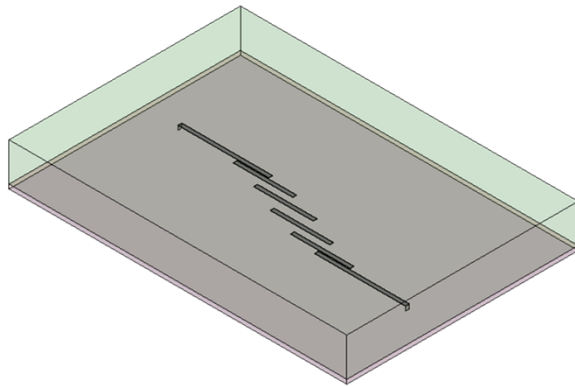


- Wave ports can be used to feed printed transmission lines
 - S-parameters normalized to computed characteristic impedance
 - Multiple propagating modes possible
 - De-embedding available as post-processing operation
 - Must touch background object (or be backed by conducting object)

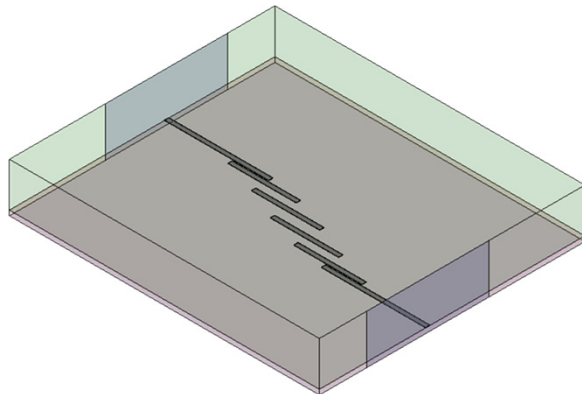


Lumped vs Wave Ports for Planar Filters

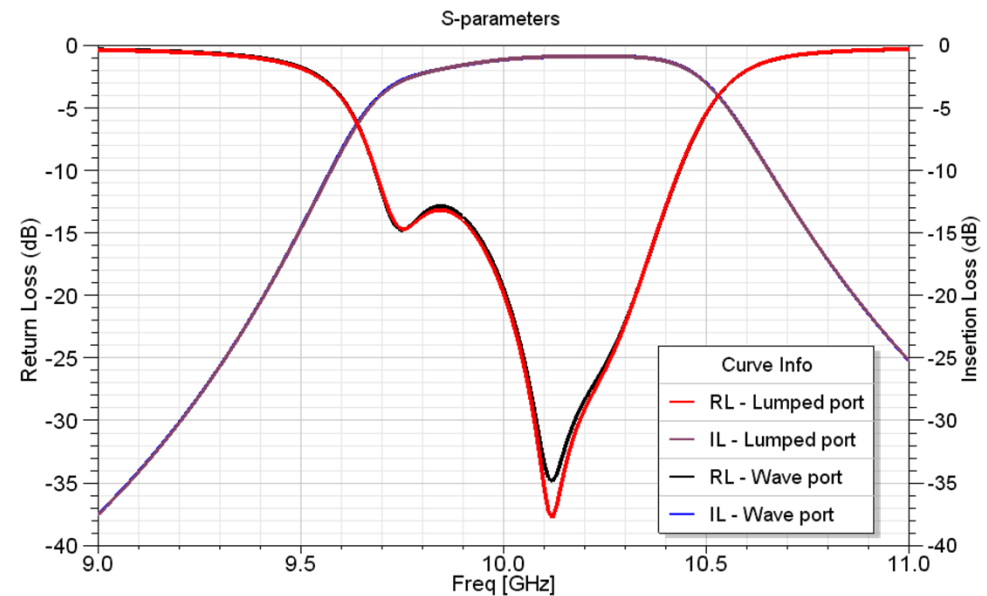
- Same results obtained from both port types



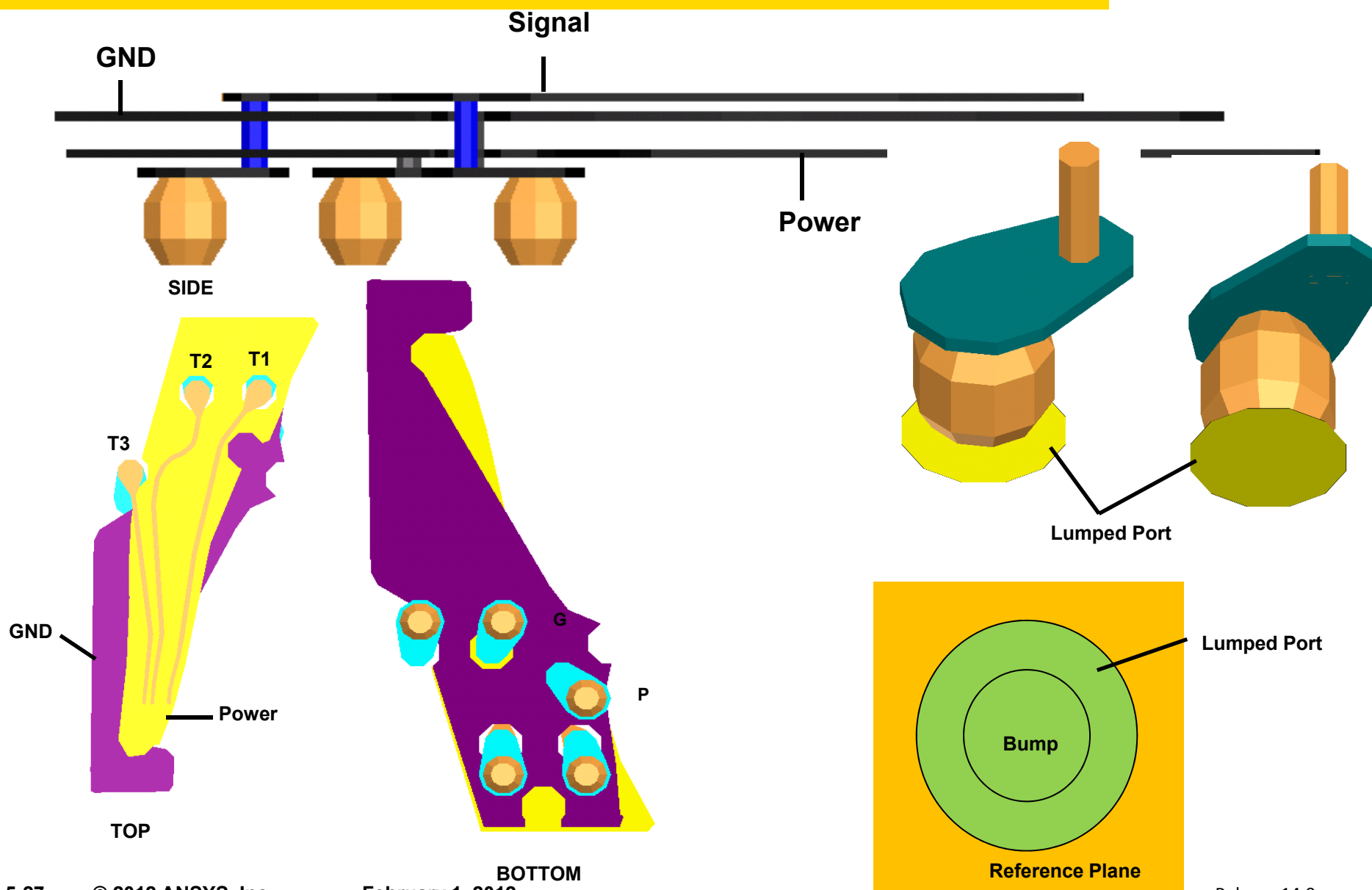
Lumped Ports



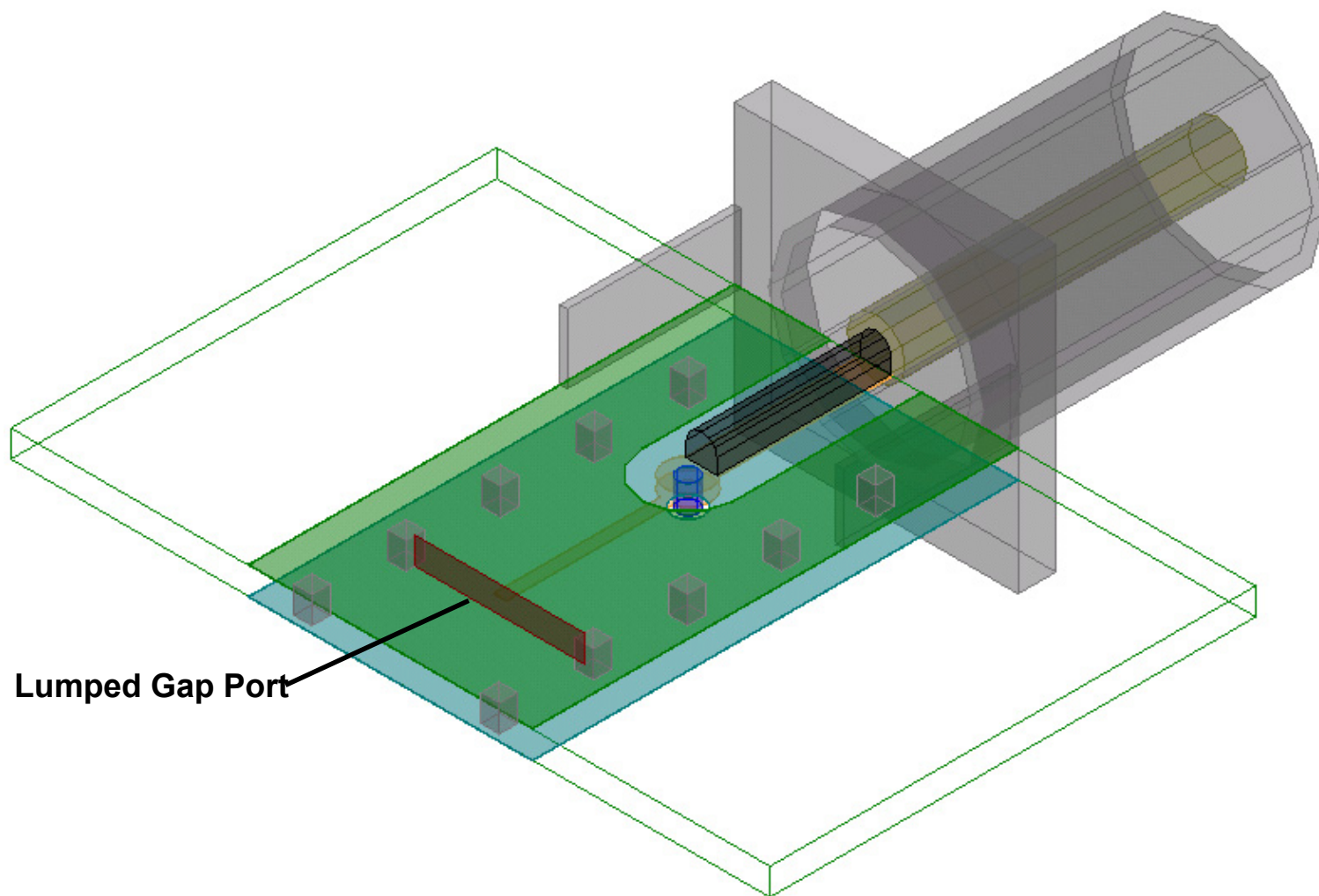
Wave Ports



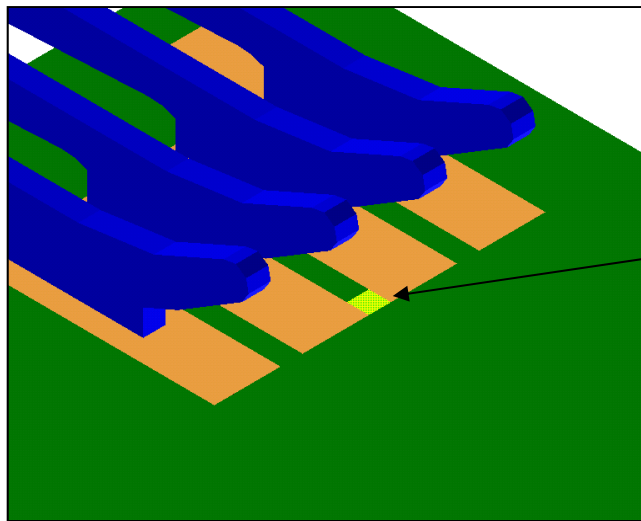
Lumped Port for Solder Balls/Bumps



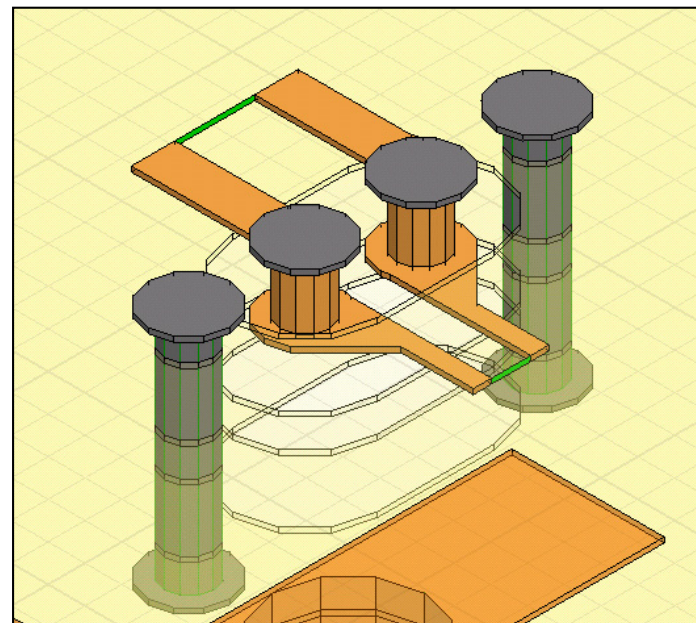
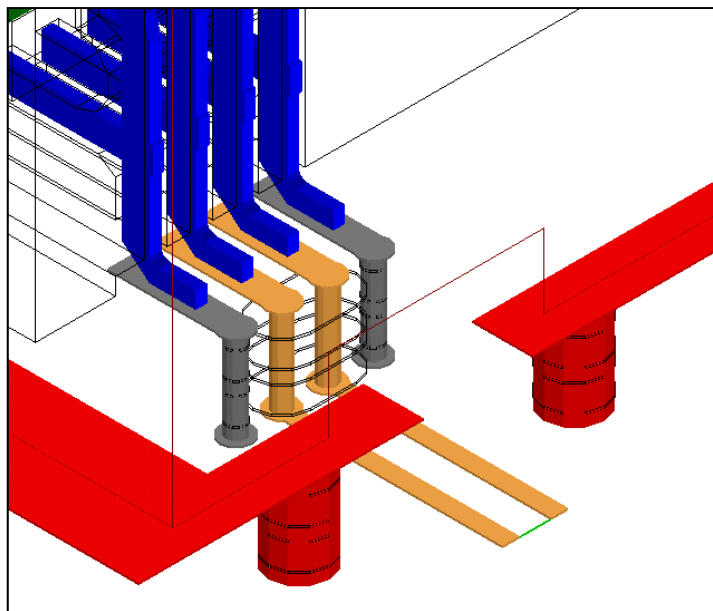
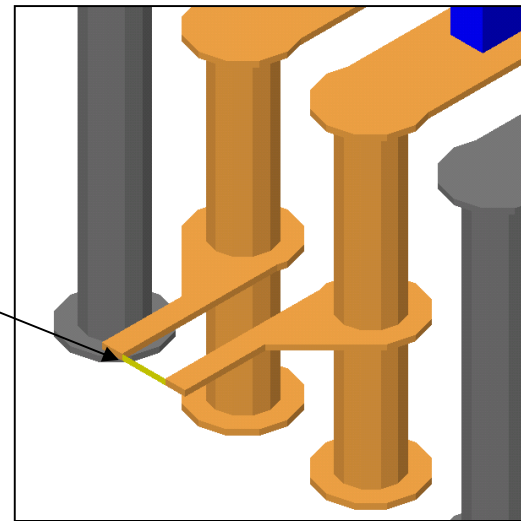
Lumped Port for Stripline



Lumped Port for Differential Mode



Differential Lumped Ports





Wave Ports vs Lumped Ports

	Wave port	Lumped port
Accessibility	External Faces	Internal to Model
Higher order modes	Yes	No
De-embedding	Yes	No
Re-normalization	Yes	Yes
Setup complexity	Moderate	Low

Summary

- **2 Types of Ports Discussed**

Wave Port

- Represents 2D Cross Section of a transmission line
- Can handle multiple modes or terminals
- Defined on planar surface or face
- Must encompass all fields that impact transmission line behavior
- Computes these TL quantities
 - Characteristic Impedance
 - Propagation Constant
 - Field Configurations

Lumped Port

- Represents a voltage source placed between conductors
- Can only handle a single TEM mode or terminal
- Defined on planar surface or face
- Must be placed between conductor
- User must specify
 - Characteristic Impedance

- **2 Solution Types Impact How Transmission Line Propagation is Interpreted**

Driven Modal

- Fields based transmission line interpretation

Driven Terminal

- Circuit Based transmission line interpretation

- **Most Important Consideration**

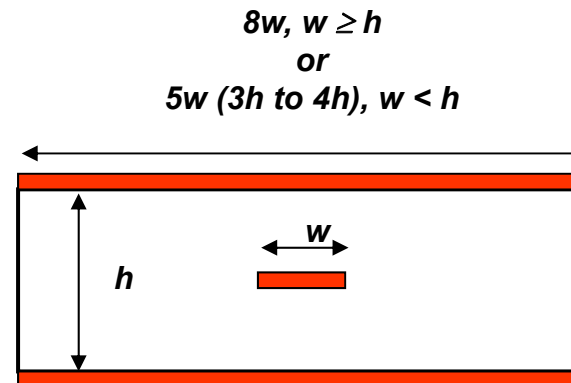
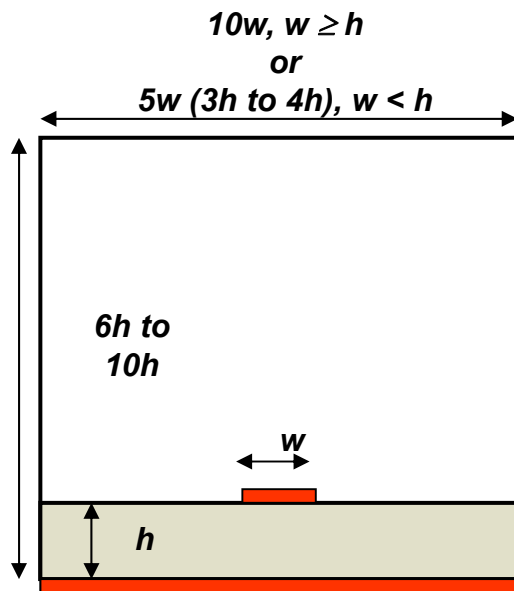
Make sure the port fields accurately reflect reality



Appendix

Wave Port Sizing Guidelines

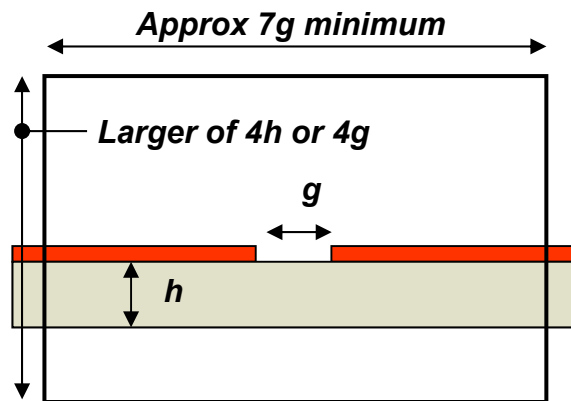
- **Microstrip port height between $6h$ and $10h$**
 - Tend towards upper limit as dielectric constant drops and fringing fields increase
 - Make bottom edge of port co-planar with upper face of ground plane
- **Microstrip port width**
 - $10w$ for $w \geq h$
 - $5w$, or on order of $3h$ to $4h$, for $w < h$
- **Extend stripline port height from upper to lower groundplane (h)**
- **Stripline port width**
 - $8w$ for $w \geq h$
 - $5w$, or on order of $3h$ to $4h$, for $w < h$
- **Can also make side walls of port Perfect H boundaries**



Port sizing guidelines are not inviolable rules. If meeting height and width requirements result in rectangular aperture larger than $\lambda/2$ in one dimension, the substrate and trace may be ignored in favor of a waveguide mode. When in doubt, run a ports-only solution to determine which modes are propagating.

Wave Port Sizing Guidelines

- **Slotline port height at least $4h$ or $4g$ (whichever is larger)**
 - Include air above and below substrate
 - If ground plane is present, port should terminate at ground plane
- **Port width should contain at least $3g$ to either side of slot or $7g$ total minimum**
 - Port boundary must intersect both side ground planes or they will 'float' and become signal conductors



For Driven Modal solutions, use Z_{pv} for impedance calculation

- **Coplanar waveguide port height at least $4h$ or $4g$ (whichever is larger)**
 - Include air above and below substrate
 - If ground plane is present, port should terminate at ground plane
- **Port width should contain $3-5g$ or $3-5s$ of side grounds (whichever is larger)**
 - Total width $\sim 10g$ or $\sim 10s$
 - Port outline must intersect both side grounds or they will 'float' and become signal conductors

